

Getting help

Today's agenda:

- Final Project
- Debugging
- AI and coding

Final Project

- Due December 11th by midnight (note: this is a Thursday)
- Includes 3 parts:
 - Code (submitted as R or .Rmd)
 - Written portion (Word document)
 - Presentation (Google slides)
- Designed to build on your previous work

Final Project: Code

- Upload to canvas or give a Github link
- Should be fully reproducible and well-documented
- Should build on what you've been doing in class
- Should essentially be a full analysis now
 - Loading data
 - Any needed transformations
 - Code needed to justify any modelling decisions
 - Model selection steps (if needed)
 - Statistical analyses
 - Figures with ggplot2

Final Project: Writing

- An updated draft of your midterm writing
- Same three sections as before
 - Introduction
 - Methods
 - Results
- Results need to include the results of a statistical model
- Figures should also be improved (e.g., using ggplot2)
- Feedback from previous draft should be incorporated

Final Project: Presentation

- Approximately 5-minute presentation
- Done during our scheduled final exam period
- Suggested format:
 - Title slide
 - Background information: 1-2 slides
 - What is your question? Why does it matter?
 - Methods: 1-2 slides
 - How did you investigate your question?
 - Results: 1-2 slides
 - Figure(s)
 - Statistical results
 - What does it mean ecologically? In terms of your question?

Final Project: Presentation

- Aim to minimize text (7x7 rule)
- Practice your timing
- Lose points for talking too long

Final Project: Questions?

Debugging

- Something you've already all had to do
- A critical part of working with code
- Lots of different techniques you'll pick up over time, e.g.:

Google

Stack Overflow

Line-by-line running

message() feedback

AI (but also a major source of bugs)

General Workflow

- 1) Re-read code for obvious errors
- 2) Run code line by line to find where things break
- 3) Google the error message
- 4) Google a description of the issue
- 5) Post on StackOverflow
- 6) Add an Issue to the package's Github page
- 7) Dig in to underlying code to fix the error yourself

Basic errors

Things to look for:

- Comma, quotation mark, and parentheses placement

- Variable names

- NAs

- Variable class (e.g., factor vs numeric)

Debugging practice

```
library(tidyverse)
```

```
library(reprex)
```

```
traits <- read_rds("https://tinyurl.com/amniotes")
```

Debugging practice

```
traits |>
```

```
  group_by(class) |>
```

```
  summarise(mean_mass = mean(adult_body_mass_g))
```

Debugging practice

```
data.frame(scientific_name = paste(traits[4,],traits[5,]))
```

Debugging practice

```
traits |>
```

```
  group_by(class) |>
```

```
  summarise(SD_sv1 = sd(adult_sv1))
```

Debugging practice

```
traits >
```

```
mutate(svl_mass_ratio = adult_svl_cm/adult_body_mass_g)
```

More complex issues

Debugging is harder when working with `for()`, `while()`, `apply()`, etc.

In these cases you can:

- Use `message()` if you need to figure out where things break
- Run line-by-line to look for errors within loops/applies
- Run only the instances that failed line-by-line
- Have R write output to figure out where things work/break

Harder debugging

```
# Calculate offspring per year, skipping those with NA data
annual_fecundity <- NULL
for(i in 1:length(unique(traits$species))){
  species_i <- paste(traits$genus[i], " ", traits$species[i], sep = " ")
  offspring_per_clutch <- traits$litter_or_clutch_size_n[i]
  clutches_per_year <- traits$litters_or_clutches_per_y[i]
  # Don't include values with NAs
  if(is.na(offspring_per_clutch) &
     is.na(clutches_per_year)){
    next
  }
  offspring_per_clutch*clutches_per_year
  out <- data.frame(species = species_i,
                    annual_offspring = offspring_per_clutch*clutches_per_year)
  annual_fecundity <- bind_rows(annual_fecundity, out)
}# end i loop
```

Harder debugging

```
# Break data into family-sized batches and save them as RDS files
```

```
# make a temp folder
```

```
family_folder <- tempdir()
```

```
# Get data for each family and save as an RDS file
```

```
for(i in 1:length(unique(traits[,4]))){
```

```
  group_i <- unique(traits[,4])[i]
```

```
  data_i <- traits[which(traits[,4] == group_i),]
```

```
  data_i |>
```

```
  saveRDS(file.path(genus_folder,
```

```
                  paste(unique(data_i[,3]), ".rds", sep = "")))
```

```
}
```

```
# Look at output files
```

```
family_files <- list.files(family_folder, pattern = ".rds", full.names = TRUE)
```

```
# Load an example
```

```
test <- readRDS(family_files[i])
```

Google

If you can't figure it out yourself, try Google

A big part of coding is Googling

- Include “R” and package names where possible
- Pasting in the error you receive can help

reprex

If you've tried everything and can't figure it out:

- 1) Ask on stackoverflow using a reproducible example (reprex)
- 2) Open an Issue on Github (also using reprex)

Stack

Stack Overflow

Stack Exchange

- Great resources

StackExchange

Search on Geographic Information Systems...

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R spTransform polygon that crosses dateline

Asked 7 years, 2 months ago Modified 7 years ago Viewed 1k times

Ask Question

▲

6

▼

I have created polygons with CRS = UTM (see [here](#)), which I need to transform to latitude/longitude. Some of my polygons cross the dateline, which causes the transformation to create a polygon that crosses the entire globe.

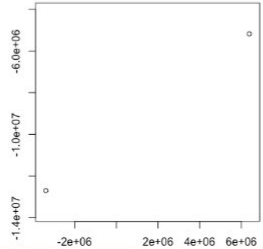
```
library(sp)
library(rgeos)

points <- data.frame(x = c(-179.5, 0), y = c(-50, -30))
coordinates(points) <- ~x+y
proj4string(points) <- '+init=epsg:4326'

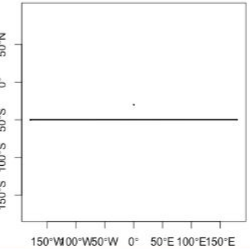
points_sp <- spTransform(points, CRS('+init=epsg:32621')) # wrong zone, but ignore
# expand point to circle
circles <- gBuffer(points_sp, width = 1e+5)
# transform back to WGS84
circles_laton <- spTransform(circles, CRS(proj4string(points)))

par(mfrow = c(1, 2))
plot(circles, axes = TRUE, main = 'UTM')
plot(circles_laton, axes = TRUE, main = 'WGS84')
```

UTM



WGS84



The Overflow Blog

Community Products roadmap update, November 2025

Only you can stop AI database drops

Featured on Meta

We're releasing our proactive anti-spam measure network-wide

Chat room owners can now establish room guidelines

Maintaining Community Value While Community Engagement Continues to Drop

Linked

14

How to prevent cross-world lines in ggplot world map?

2

R Raster: extract weighted mean within circle of specific radius

Related

7

Can extracting raster values work in polygons' CRS without raster resampling?

1

Why did spTransform fail here?

0

spTransform produces unexpected results

2

Extracting pixel values in different bands for different points giving NAs and not pixel values

2

Using spatial points to extract values from large raster in R

Acti
Go to

21

Github issues

Place to note bugs

- Use reprex

EnquistLab / RGVs

Code Issues Pull requests Actions Projects Security Insights Settings

RGVs Public

Edit Pins Unwatch 13 Fork 0 Star 0

main

Go to file

Code

bmainer incremented version for dev 6a8285e · last year 23 Commits

.github/workflows	rhub	last year
R	update after minor change to exa...	last year
data	added data, updated base URL	last year
man	update after minor change to exa...	last year
old_builds	update after minor change to exa...	last year

About

R package to access the GVS from BIEN

Readme

Unknown, MIT licenses found

Activity

Custom properties

0 stars

13 watching

0 forks

Activate Windows
Go to Settings to activate Wind

Report repository

reprex

R package reprex

- Helps you make an easily reproducible example
- If your example is not reproducible, people will be jerks to you online
- (If your example is reproducible, people may still be jerks)

reprex

```
reprex({  
  
  traits <- read_rds("https://tinyurl.com/amniotes")  
  
  traits |>  
  ggplot(mapping = aes(x=adult_body_mass_g,  
                        y=adult_svl_cm))+  
    geom_point()+  
    scale_y_log10()+  
    scale_x_log10()+  
    geom_smooth(method = "lm")  
  
})
```

Run this and examine the output.

reprex

Make a version without the error message

```
traits <-  
read_rds("https://tinyurl.com/amniotes")  
#> Error in  
read_rds("https://tinyurl.com/amniotes")  
: could not find function "read_rds"
```

```
traits |>  
ggplot(mapping =  
aes(x=adult_body_mass_g,  
      y=adult_svl_cm))+  
  geom_point()+  
  scale_y_log10()+  
  scale_x_log10()+  
  geom_smooth(method = "lm")  
#> Error in ggplot(traits, mapping =  
aes(x = adult_body_mass_g, y =  
adult_svl_cm)): could not find  
function "ggplot"
```

Asking for help

When asking for help online or reporting an issue:

- Be very nice
- Make sure you supply a well made repro
- Remember that you're asking for a favor
- Assume the issue is user error on your end

Help from AI

One newer option is to ask AI for help (e.g., ChatGPT)

ChatGPT can be used for:

- Writing bits of code
- Finding packages to use
- Summarizing what code is doing
- Debugging

Issues with using AI

- The training information may be outdated
- Often hallucinates packages, functions
- Will confidently give you wrong answers
- You still need to be able to debug
- Issues with sustainability of AI data centers
- Issues with proper crediting of people writing code
- Ultimately you still need to justify your decisions based on the literature
 - E.g., “ChatGPT said to do it this way” isn’t acceptable

Let's try ChatGPT

If you have ChatGPT, try giving it some code with an error in it.

- Can it identify what the error is?
- Can it suggest a solution?
- Does the solution actually work?

Take-away: General debugging workflow

- 1) Re-read code for obvious errors
- 2) Run code line by line to find where things break
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Before next class:

- Read “R for Data Science” chapter 6:
 - <https://r4ds.hadley.nz/workflow-scripts.html>
- May be a quiz